

# Muhammad Ayan Butt

## Machine Learning Engineer | Scientific ML, Deep Learning, LLMs

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ML engineer with two years of research at CESAT building deep learning surrogates for aerodynamic design, validated against CFD and wind tunnel data. Also implemented GPT-2 from scratch in PyTorch and built a leakage-audited content decay model over 148K URLs at FlyRank AI.

## EXPERIENCE

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### Research Assistant, AI & Aerodynamics · CESAT

Sept. 2024 – Present · Islamabad, Pakistan

- Co-authored "Airfoil Shape Optimization for Enhanced Aerodynamic Efficiency Using a Dual-Branch Deep Network Integrated with Genetic Algorithms", under review at an international journal.
- Designed and trained a dual-branch deep neural network in PyTorch that predicts aerodynamic coefficients from airfoil geometry and flow conditions, replacing per-candidate CFD solves.
- Coupled the network with a genetic algorithm that searches the airfoil design space, converging faster than gradient-based optimization.
- Validated optimized airfoils against ANSYS Fluent simulations and subsonic wind tunnel tests.

### Machine Learning Intern · FlyRank AI

Jun. 2026 – Jul. 2026 · USA (Remote) · [Project write-up](#)

- Framed content decay as binary classification over 148,133 URLs across 57 client domains, drawn from a warehouse of 79M+ impression events.
- Trained a Random Forest under GroupKFold by client (0.503 PR-AUC on unseen domains against 0.481 for the rule baseline), and showed naive K-fold overstated PR-AUC by 0.097 through client-level leakage.
- Turned model probabilities into a reason-coded playbook sorting all 148K items into five editorial action categories, with human-review guardrails.

## PROJECTS

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### GPT-2 from Scratch with Instruction Fine-Tuning · PyTorch

- Implemented the GPT-2 architecture from scratch in PyTorch and loaded OpenAI's pretrained 355M weights into it.
- Fine-tuned the model on instruction-following data with supervised fine-tuning to produce a working chatbot.

### Physics-Informed Neural Network for the 2D Poisson Equation · PyTorch

- Built a PINN from scratch that solves the 2D Poisson equation by training on the PDE residual and boundary conditions rather than labeled data.

### Local RAG Chatbot over the Mechanical Engineers Handbook · LangChain, Ollama, Llama 3

- Built a fully local question-answering chatbot using Llama 3 and Nomic embeddings served through Ollama, with no external API calls.

## EDUCATION

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### B.S. in Mechanical Engineering · Pakistan Institute of Engineering and Applied Sciences (PIEAS)

Jul. 2024 · Islamabad, Pakistan

- CGPA 3.87/4.00, ranked 2nd of 46. High-Achiever Scholarship and Research Fellowship.
- Thesis: Design of High-Performance Aero-Vehicle System using Integrated CFD, ML, and Systems Engineering Techniques.

## SKILLS

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- **ML:** Python, PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, YOLO
- **LLMs and deployment:** LangChain, FAISS, ChromaDB, Ollama, FastAPI, ONNX, n8n, Git, Bash
- **Simulation:** ANSYS Fluent, SolidWorks (API and macros), MATLAB, C/C++
- **Certifications and languages:** PyTorch and TensorFlow for Deep Learning (Coursera); English (IELTS 7.5), Urdu (native)

## LEADERSHIP

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- **Head of Workshops, ASME Chapter, PIEAS** (Oct. 2022 – May 2023): organized ANSYS, MATLAB and SolidWorks workshops with 60+ participants each.